This note describes the problem with a lender of last resort (LOLR). We assume one-period debt. Notation is very similar to the previous note.

I describe two alternative models. Section 1 has an LOLR akin to the policies the IMF implements. Section 2 describes a case that resembles the ECB type of policy.

1 LOLR Model: IMF version

Value of no default.

$$
\begin{aligned}
v^{nd}(b,\ell,g,s,\epsilon) &= \max_{c,n,\ell'}\left\{\frac{c^{1-\gamma}}{1-\gamma} + \beta g^{1-\gamma}\mathbb{E}\left[\max\left\{v^{nd}(b',\ell',g',s',\epsilon'), v^{d}(b',\ell',g',s',\epsilon')\right\}|g,s\right]\right\} \qquad (1)\\
c+b+\ell &= y+n+n^{\ell}\\
y &= ge^{\sigma\epsilon}\\
gb' &= R(n,\ell',g,s)n, \quad g\ell' = R^{\ell}_{nd}n^{\ell}\\
\ell' &\le \bar{\ell}_{nd}, \quad n\le \bar{n}(g)
\end{aligned}
$$

where ℓ is the amount of debt outstanding with the LOLR, and n^{ℓ} is the new issuance with the LOLR. We assume the LOLR charges a given (gross) rate R^{ℓ}_{nd}. There is a limit $\bar{\ell}_{nd}$ on how much the sovereign can borrow from the LOLR. The LOLR rate and borrowing limit depend on default status but not on other states—we'll make a function of s later on.

We are assuming that the schedule from private lenders depends on ℓ: that is, $R(n,\ell',g,s)$ This implies a timing assumption: the sovereign borrows from the LOLR first, and then borrows from private markets. [LETS DISCUSS ABOUT THIS]

Value of default

$$
\begin{aligned}
v^{d}(b,\ell,g,s,\epsilon) &= \max_{c,n^{\ell}}\left\{\frac{c^{1-\gamma}}{1-\gamma} + \beta g^{1-\gamma}\mathbb{E}\Big[\theta v^{d}(b',\ell',g',s',\epsilon')\right. \qquad (2)\\
&\quad \left. + (1-\theta)\max\left\{v^{nd}(\kappa b',\ell',g',s',\epsilon'), v^{d}(b',\ell',g',s',\epsilon')\right\}|g\Big]\right\}\\
c+\ell &= \phi(g)y+n^{\ell}\\
y &= ge^{\sigma\epsilon}\\
gb' &= b, \quad g\ell' = R^{\ell}_{d}n^{\ell}\\
\ell' &\le \bar{\ell}_{d}
\end{aligned}
$$

The assumption is that the sovereign cannot default on the LOLR. We further assume that the sovereign can also keep on borrowing from the LOLR during periods of default/market exclusion. However, the LOLR can set a rate R^{ℓ}_{d} and a borrowing limit $\bar{\ell}_{d}$ that are different during periods of default/market exclusion.

Let $\boldsymbol{b}'(b,\ell,g,s,\epsilon)$ be the optimal borrowing policy, $\boldsymbol{\ell}'_{nd}(b,\ell,g,s,\epsilon)$ be the borrowing policy from

LOLR during no default, and $\boldsymbol{\ell}'_d(b,\ell,g,s,\epsilon)$ be the borrowing policy from LOLR during default.

Schedule

$$1 = \frac{R(n,\ell',g,s)}{1+r^*}\mathbb{E}[Q(b',\ell',g',s'\epsilon')|g,s] \tag{3}$$

$$gb' = R(n,\ell',g,s)n \tag{4}$$

with $\ell' = \boldsymbol{\ell}'_{nd}(b,\ell,g,s,\epsilon)$ in equation (4).

Let $z = \{g,s,\epsilon\}$ collect all the exogenous sates. Prices are given as

$$Q(b,\ell,z) = [1-\boldsymbol{d}(b,\ell,z)] + \boldsymbol{d}(b,\ell,g,\epsilon)X(b,\ell,z) \tag{5}$$

$$\begin{aligned} X(b,\ell,z) = \frac{1}{1+r^*}\mathbb{E}\Big[&\theta X(b',\boldsymbol{\ell}'_d(b,\ell,z),z') + (1-\theta)\times \\ &\left\{\left[1-\boldsymbol{e}(b',\boldsymbol{\ell}'_d(b,\ell,z),z')\right]X(b',\boldsymbol{\ell}'_d(b,\ell,z),z') + \boldsymbol{e}(b',\boldsymbol{\ell}'_d(b,\ell,z),z')\kappa Q(\kappa b',\boldsymbol{\ell}'_d(b,\ell,z),z')\right\}|g\Big] \end{aligned} \tag{6}$$

where $\boldsymbol{d}(\cdot)$ and $\boldsymbol{e}(\cdot)$ are the defaul and re-entry policies, and $b' = b/g$ in equation (6). IMPORTANT: to solve equation (6) we need the borrowing policies: $\boldsymbol{b}'(b,\ell,z)$, $\boldsymbol{\ell}'_{nd}(b,\ell,z)$, $\boldsymbol{\ell}'_d(b,\ell,z)$. The borrowing policy $\boldsymbol{b}'(b,\ell,z)$ is implied by the optimal issuance policy and the schedule as in equation (1).

The default and re-entry decisions, $\boldsymbol{d}(b,\ell,g,s,\epsilon)$ and $\boldsymbol{e}(b,\ell,g,s,\epsilon)$, are given as

$$\boldsymbol{d}(b,\ell,g,s,\epsilon) = \begin{cases} 0, & \text{if } v^{nd}(b,\ell,g,s,\epsilon) \geq v^d(b,\ell,g,s,\epsilon), \\ 1, & \text{otherwise;} \end{cases} \tag{7}$$

$$\boldsymbol{e}(b,\ell,g,s,\epsilon) = \begin{cases} 1, & \text{if } v^{nd}(b,\ell,g,s,\epsilon) \geq v^d(b,\ell,g,s,\epsilon), \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

1.1 Implementation Comments

We need to set the borrowing limits $\left\{\bar{\ell}_{nd}, \bar{\ell}_d\right\}$ and the rates $\left\{R^\ell_{nd}, R^\ell_d\right\}$. Let's start assuming the borrowing limit and the rates do not depend on the market status: $\bar{\ell}_{nd} = \bar{\ell}_d = \bar{\ell}$ and $R^\ell_{nd} = R^\ell_{nd} = R^\ell$.

For the borrowing limit. Let $\bar{g} = \mathbb{E}[g]$ be the average growth rate (on the ergodic distribution). I would set $\bar{\ell} = \Delta\bar{g}$. Start with small numbers, moving from 1% to 5%. That is $\Delta \in \{0.01,\ 0.02,\ 0.03,\ 0.04,\ 0.05\}$.

For the interest rate. Take the risk-free rate and add a few points. Say from 4 p.p. to 10 p.p. So we wold have $R^\ell = 1 + r^* + \sigma^\ell$ and $\sigma^\ell = \left\{\frac{4}{100},\ \frac{6}{100},\ \frac{8}{100},\ \frac{10}{100}\right\}$. We can explore more values for σ^ℓ later on. Ideally, we want to make sure σ^ℓ is large enough so that it's not subsidizing borrowing.

1.2 Solution algorithm

Step 0: Set grids. Grid for debt, $\vec{b} = \{b_1, b_2, \ldots, b_{N_b}\}$, with $b_1 < 0$ (saving) and $b_{N_b} > 0$ (borrowing). Grid for LOLR lending $\vec{\ell} = \{\ell_1, \ell_2, \ldots, \ell_{N_\ell}\}$ Grid for endowment shocks, $\vec{\epsilon} = \{\epsilon_1, \epsilon_2, \ldots, \epsilon_{N_\epsilon}\}$.

I would start with small values for N_b and N_ℓ. Probably around 1000 for each. Once it's running, we will want to increase the grid sizes.

For $\vec{\epsilon}$ let's do the same as in the paper: $N_\epsilon = 17$ and use Tauchen method. (We should move to Rouwenhorst method. See here: https://www.karenkopecky.net/RouwenhorstPaper.pdf).

We also need $\vec{g} = \{g_L, g_H\}$, the transition matrix $P_g(g'|g)$, and the variance σ_ϵ. All these numbers come from the estimation (see Table 1 in our paper).

Step 1: Guess $v^{nd}(b, \ell, g, s, \epsilon)$. Also guess borrowing policies from the LOLR during no default: $\boldsymbol{\ell}'_{nd}(b, \ell, g, s, \epsilon)$.

Step 2: Given $v^{nd}(b, \ell, g, s, \epsilon)$, solve for $v^d(b, \ell, g, s, \epsilon)$. Note that, given $v^{nd}(b, \ell, g, s, \epsilon)$, equation (2) is a fixed point in $v^d(b, \ell, g, s, \epsilon)$. Iterate on equation (2) until $v^d(\cdot)$ convergences.

From this step, save $\boldsymbol{\ell}'_d(b, \ell, g, s, \epsilon)$

Step 3: Given $v^{nd}(b, \ell, g, s, \epsilon)$ and $v^d(b, \ell, g, s, \epsilon)$, compute $\boldsymbol{d}(b, \ell, g, s, \epsilon)$ and $\boldsymbol{e}(b, \ell, g, s, \epsilon)$ as in equations (7)-(8).

Step 4: Given the borrowing policy $\boldsymbol{\ell}'_d(b, \ell, g, s, \epsilon)$, the default and re-entry decisions, $\boldsymbol{d}(b, \ell, g, s, \epsilon)$ and $\boldsymbol{e}(b, \ell, g, s, \epsilon)$, and our guess for borrowing policies $\boldsymbol{\ell}'_{nd}(b, \ell, g, s, \epsilon)$, we solve for the prices $Q(b, \ell, g, s, \epsilon)$ and $X(b, \ell, g, s, \epsilon)$ using equations (5)-(6).

In particular, guess $X(b, g, , s, \epsilon)$. Then compute the price $Q(b, g, s, \epsilon)$ using equation (5), and the implied $X(b, g, \epsilon)$ using equation (6). Iterate until $X(\cdot)$ converge.

Step 5: We need to compute the schedule $R(n, \ell', g, s)$. This is the hardest step, and even more now with two debt choices. The computations below is for a given g.

Consider debt levels for next period: $b' = b_i$ and $\ell' = \ell_j$. That is, private debt is the i^{th} position of the grid $\vec{b}$ and LOLR debt is the j^{th} position of the grid $\vec{\ell}$.

Compute the expected price tomorrow associated with these decisions $b' = b_i$ and $\ell' = \ell_j$. That is: $Q^{\mathbb{E}}_{ij}(g, s) = \mathbb{E}[Q(b_i, \ell_j, g', s', \epsilon')|g, s]$. The implied rate then is $R_{ij}(g, s) = \frac{1+r^*}{Q^{\mathbb{E}}_{ij}(g,s)}$. The issuance then is $n_{ij}(g, s) = \frac{g b_i}{R_{ij}(g,s)}$

We can do this $\forall i = 1, \ldots, N_b$ and $\forall j = 1, \ldots, N_\ell$. The implied schedule is a collection of points as $\tilde{\mathcal{S}}(g, s) = \{R_{ij}(g, s), n_{ij}(g, s)\}_{i,j}$.

Compute expected default probability $p_{ij}(g, s) = \mathbb{E}[\mathbf{d}(b_i, \ell_j, g', s', \epsilon')|g, s]$. We will impose an upper bound p_{ub} on schedule choices based on this default probability. In particular, take the points of the schedule with default probability below p_{ub}: $\mathcal{S}(g, s) = \left\{ \{R_{ij}(g, s), n_{ij}(g, s)\}_{ij} : p_{ij}(g, s) \leq p_{ub} \right\}$.

Step 6: We need to determine which parts of the schedule are available to the borrower depending on the sunspot realization, s.

In the good sunspot, $s = s_g$, we allow the borrower to choose from all of the schedule $\mathcal{S}(g, s_g)$. (Keep in mind that we already eliminated the high-default probability choices).

In the bad sunspot, $s = s_b$, we select points with two criteria: (1) has to be an increasing part of the schedule, and (2) for a given issuance, if there is a higher rate in an increasing part of the schedule, we jump to the higher rate.

Step 7: Now we update $v^{nd}(b, \ell, g, s, \epsilon)$ and $\boldsymbol{\ell}'_{nd}(b, \ell, g, s, \epsilon)$. For given current states $\{b, \ell, g, s, \epsilon\}$,we do as follows.

Given our current guess $v^{nd}(\cdot)$, compute the value $w(b, \ell, g, s, \epsilon) = \max\left\{v^{nd}(b, \ell, g, s, \epsilon), v^{d}(b, \ell, g, s, \epsilon)\right\}$, and the expected value $w^{\mathbb{E}}(b, \ell, s, g) = \mathbb{E}\left[w(b, \ell, g', s', \epsilon')|g, s\right]$.

That is, for each $b' = b_i \in \vec{b}$ and $\ell' = \ell_j \in \vec{\ell}$, evaluate the value of the policy

$$
\begin{aligned}
\hat{v}^{nd}_{ij}(b, \ell, g, s, \epsilon) &= \frac{c_{ij}^{1-\gamma}}{1-\gamma} + \beta g^{1-\gamma} w^{\mathbb{E}}(b_i, \ell_j, g, s) \qquad (9)\\
c_{ij} &= g e^{\sigma\epsilon} + n_{ij}(g, s) + n^{\ell}_j - b - \ell \\
n^{\ell}_j &= \frac{g\ell_j}{R^{\ell}}
\end{aligned}
$$

Then, we update our guess

$$
\hat{v}^{nd}(b, \ell, g, s, \epsilon) = \max_{i,j}\left\{\hat{v}^{nd}_{ij}(b, \ell, g, \epsilon) \;\text{ s. to }\; \ell_j \le \bar{\ell} \;\text{ and }\; n_{ij} \in \mathcal{S}(g, s)\right\} \qquad (10)
$$

When computing the max in equation (10), we need to impose that the issuance choice is within the schedule that corresponds to the sunspot s. In practice, when $s = s_b$, we impute a very negative pay-off for choices that are not in the schedule.

This optimization will also give a new borrowing policy from the LOLR: $\hat{\boldsymbol{\ell}}'(b, \ell, g, s, \epsilon) = \ell_j$.

Compare $\hat{v}^{nd}(b, \ell, g, s, \epsilon)$ and $\hat{\boldsymbol{\ell}}'(b, \ell, g, s, \epsilon)$ to the initial guess $v^{nd}(b, \ell, g, s, \epsilon)$ and $\boldsymbol{\ell}'(b, \ell, g, s, \epsilon)$. If it's close, we are done. Otherwise, update $v^{nd}(b, \ell, g, s, \epsilon)$ and $\boldsymbol{\ell}'(b, \ell, g, s, \epsilon)$, and go back to *Step 2*.

2 LOLR Model: ECB version

In this case, we assume that the LOLR doesn't lend directly to the sovereign. Instead, the LOLR intervenes in the private credit market by buying/selling bonds directly from private lenders.

Sovereign We describe the problem of the sovereign next.

Value of no default.

$$v^{nd}(b,g,s,\epsilon) = \max_{c,n} \left\{ \frac{c^{1-\gamma}}{1-\gamma} + \beta g^{1-\gamma} \mathbb{E}\left[\max\left\{ v^{nd}(b',g',s',\epsilon'), v^{d}(b',g',s',\epsilon') \right\} |g,s \right] \right\} \tag{11}$$

$$\begin{aligned} c + b &= y + n \\ y &= g e^{\sigma\epsilon} \\ g b' &= R(n,g,s) n, \\ n &\leq \bar{n}(g) \end{aligned}$$

Note there is no ℓ in this set-up: the sovereign doesn't borrow directly from the LOLR.

Value of default

$$\begin{aligned} v^{d}(b,g,s,\epsilon) = \max_{c,n^{\ell}} \Bigg\{ & \frac{c^{1-\gamma}}{1-\gamma} + \beta g^{1-\gamma} \mathbb{E}\Big[\theta v^{d}(b',g',s',\epsilon') \\ & + (1-\theta) \max\left\{ v^{nd}(\kappa b',g',s',\epsilon'), v^{d}(b',g',s',\epsilon') \right\} |g \Big] \Bigg\} \end{aligned} \tag{12}$$

$$\begin{aligned} c + \ell &= \phi(g) y \\ y &= g e^{\sigma\epsilon} \\ g b' &= b, \end{aligned}$$

Again, no LOLR lending in no-default state.

LOLR We assume that the LOLR has a facility that buys bonds from private creditors at a price $\mathcal{P}(d,g,s)$ up to a limit $\bar{\ell}$. The price functions $\mathcal{P}(d,g,s)$ and the borrowing limit $\bar{\ell}$ are policies of the LOLR. (Notation: d is default status.)

The timing is as follows. First, private lenders and sovereign market clears. Immediately after that, the lenders can sell the bond to the LOLR, or keep it until next period.

If there more bonds than $\bar{\ell}$ show up at the facility, the LOLR buys them randomly. That is, if n bonds show up at the facility, the probability of selling them is $\pi(n) = \min\left\{\frac{\bar{\ell}}{n}, 1\right\}$. [Note: I'm using n as in the total sovereign issuance. The reason is that, if one lenders wants to sell at the LOLR facility, all of them will want to]

Schedule Lets work out the bond price with the intervention

Let $\boldsymbol{b}'(b,g,s,\epsilon)$ be the optimal borrowing policy.

Let $\mathcal{H}(b, g, s)$ be the expected return to a private lender of keeping the bond.

$$\mathcal{H}(b', g, s) = \frac{\mathbb{E}\left[Q(b', g', s', \epsilon')|g, s\right]}{1 + r^*} \tag{13}$$

where $Q(\cdot)$ is the price of the bond next period (discussed above).

Let $\mathcal{L}(\cdot)$ be the expected return of queuing up at the LOLR facility. Then, the schedule

$$\begin{aligned} 1 &= R(n, g, s) \max\left\{\mathcal{L}(b', g, s), \frac{\mathbb{E}\left[Q(b', g', s', \epsilon')|g, s\right]}{1 + r^*}\right\} && (14) \\ \mathcal{L}(b', g, s) &= \pi(n)\mathcal{P}(d, g, s) + (1 - \pi(n))\frac{\mathbb{E}\left[Q(b', g', s', \epsilon')|g, s\right]}{1 + r^*} && (15) \\ gb' &= R(n, g, s)n && (16) \end{aligned}$$

Let $z = \{g, s, \epsilon\}$ collect all the exogenous sates. Prices are given as

$$\begin{aligned} Q(b, z) &= [1 - \boldsymbol{d}(b, z)] + \boldsymbol{d}(b, g, \epsilon)X(b, z) && (17) \\ X(b, z) &= \frac{1}{1 + r^*}\mathbb{E}\Big[\theta X(b', z') + (1 - \theta)\times && (18) \\ &\qquad \left\{\left[1 - \boldsymbol{e}(b', z')\right] X(b', z') + \boldsymbol{e}(b', z')\kappa Q(\kappa b', z')\right\} |g\Big] \end{aligned}$$

where $\boldsymbol{d}(\cdot)$ and $\boldsymbol{e}(\cdot)$ are the default and re-entry policies, and $b' = b/g$ in equation (6). IMPORTANT: to solve equation (6) we need the borrowing policies: $\boldsymbol{b'}(b, \ell, z)$ which is implied by the optimal issuance policy and the schedule as in equation (1).

The default and re-entry decisions, $\boldsymbol{d}(b, g, s, \epsilon)$ and $\boldsymbol{e}(b, g, s, \epsilon)$, are given as

$$\boldsymbol{d}(b, g, s, \epsilon) = \begin{cases} 0, & \text{if } v^{nd}(b, g, s, \epsilon) \geq v^{d}(b, g, s, \epsilon), \\ 1, & \text{otherwise;} \end{cases} \tag{19}$$

$$\boldsymbol{e}(b, g, s, \epsilon) = \begin{cases} 1, & \text{if } v^{nd}(b, g, s, \epsilon) \geq v^{d}(b, g, s, \epsilon), \\ 0, & \text{otherwise.} \end{cases} \tag{20}$$

2.1 Implementation Comments

We need to set $\mathcal{P}(d, g, s)$ and the borrowing limit $\bar{\ell}$.

Let's start with the following. When in no-default, in low growth, and the bad sunspot, let's set $\mathcal{P} = 1/R^{\ell}$. In any other state, set $\mathcal{P} = 0$. For R^{ℓ} do the same cases as before: $\sigma^{\ell} = \left\{\frac{4}{100}, \frac{6}{100}, \frac{8}{100}, \frac{10}{100}\right\}$. Ideally, we want to make sure σ^{ℓ} is large enough so that it's not subsidizing borrowing.

For the borrowing limit. Let's do something easier: assume a probability of intervention, π, that is independent of the issuance level n. Use values $\pi \in \{0, 0.25, 0.50, 0.75, 1\}$. [Set $\pi \neq 0$ only for: low-growth, bad sunspot, and no default].

[EVENTUALLY, WE'LL DO THIS: Let $\bar{g} = \mathbb{E}[g]$ be the average growth rate (on the ergodic distribution). I would set $\bar{\ell} = \Delta\bar{g}$. Start with small numbers, moving from 1% to 5%. That is $\Delta \in \{0.01,\ 0.05,\ 0.10,\ 0.20,\ 0.30\}$.]

2.2 Solution algorithm

Step 0: Set grids. Grid for debt, $\vec{b} = \{b_1, b_2, \ldots, b_{N_b}\}$, with $b_1 < 0$ (saving) and $b_{N_b} > 0$ (borrowing). Grid for endowment shocks, $\vec{\epsilon} = \{\epsilon_1, \epsilon_2, \ldots, \epsilon_{N_\epsilon}\}$.

I would start with small values for N_b, probably around 1500. Once it's running, we will want to increase the grid sizes.

For $\vec{\epsilon}$ let's do the same as in the paper: $N_\epsilon = 17$ and use Tauchen method. (We should move to Rouwenhorst method. See here: https://www.karenkopecky.net/RouwenhorstPaper.pdf).

We also need $\vec{g} = \{g_L, g_H\}$, the transition matrix $P_g(g'|g)$, and the variance σ_ϵ. All these numbers come from the estimation (see Table 1 in our paper).

Step 1: Guess $v^{nd}(b, g, s, \epsilon)$.

Step 2: Given $v^{nd}(b, g, s, \epsilon)$, solve for $v^d(b, g, s, \epsilon)$. Note that, given $v^{nd}(b, g, s, \epsilon)$, equation (12) is a fixed point in $v^d(b, g, s, \epsilon)$. Iterate on equation (12) until $v^d(\cdot)$ convergences.

Step 3: Given $v^{nd}(b, g, s, \epsilon)$ and $v^d(b, g, s, \epsilon)$, compute $\boldsymbol{d}(b, g, s, \epsilon)$ and $\boldsymbol{e}(b, g, s, \epsilon)$ as in equations (19)-(20).

Step 4: Given the default and re-entry decisions, $\boldsymbol{d}(b, g, s, \epsilon)$ and $\boldsymbol{e}(b, g, s, \epsilon)$, we solve for the prices $Q(b, g, s, \epsilon)$ and $X(b, g, s, \epsilon)$ using equations (17)-(18).

In particular, guess $X(b, g, s, \epsilon)$. Then compute the price $Q(b, g, s, \epsilon)$ using equation (17), and the implied $X(b, g, \epsilon)$ using equation (18). Iterate until $X(\cdot)$ converge.

Step 5: We need to compute the schedule $R(n, g, s)$. This is the hardest step, and even more now with two debt choices. The computations below is for a given g.

Consider debt levels for next period: $b' = b_i$. That is, private debt is the i^{th} position of the grid $\vec{b}$.

Compute the expected price tomorrow associated with these decisions $b' = b_i$. That is: $Q_i^{\mathbb{E}}(g, s) = \mathbb{E}[Q(b_i, g', s', \epsilon')|g, s]$. Then, compute the LOLR facility value: $\mathcal{L}_i(g, s) = \pi\mathcal{P} + (1 - \pi)\frac{Q_i^{\mathbb{E}}(g,s)}{1+r^*}$. Let $\mathcal{M}_i(g, s) = \max\left\{\mathcal{L}_i(g, s), \frac{Q_i^{\mathbb{E}}(g,s)}{1+r^*}\right\}$

The implied rate as: $R_i(g, s) = 1/\mathcal{M}_i(g, s)$. The issuance then is $n_i(g, s) = \frac{gb_i}{R_i(g,s)}$

We can do this $\forall i = 1, \ldots, N_b$. The implied schedule is a collection of points as $\tilde{\mathcal{S}}(g, s) = \{R_i(g, s), n_i(g, s)\}_i$.

Compute expected default probability $p_i(g, s) = \mathbb{E}[\mathbf{d}(b_i, g', s', \epsilon')|g, s]$. We will impose an upper bound p_{ub} on schedule choices based on this default probability. In particular, take the points of the schedule with default probability below p_{ub}: $\mathcal{S}(g, s) = \{\{R_i(g, s), n_i(g, s)\}_i : p_i(g, s) \leq p_{ub}\}$.

Step 6: We need to determine which parts of the schedule are available to the borrower depending on the sunspot realization, s.

In the good sunspot, $s = s_g$, we allow the borrower to choose from all of the schedule $\mathcal{S}(g, s_g)$. (Keep in mind that we already eliminated the high-default probability choices).

In the bad sunspot, $s = s_b$, we select points with two criteria: (1) has to be an increasing part of the schedule, and (2) for a given issuance, if there is a higher rate in an increasing part of the schedule, we jump to the higher rate.

Step 7: Now we update $v^{nd}(b, g, s, \epsilon)$. For given current states $\{b, g, s, \epsilon\}$,we do as follows.

Given our current guess $v^{nd}(\cdot)$, compute the value $w(b, g, s, \epsilon) = \max\left\{v^{nd}(b, g, s, \epsilon), v^d(b, g, s, \epsilon)\right\}$, and the expected value $w^{\mathbb{E}}(b, s, g) = \mathbb{E}\left[w(b, g', s', \epsilon')|g, s\right]$.

That is, for each $b' = b_i \in \vec{b}$, evaluate the value of the policy

$$\hat{v}_i^{nd}(b, g, s, \epsilon) = \frac{c_i^{1-\gamma}}{1-\gamma} + \beta g^{1-\gamma} w^{\mathbb{E}}(b_i, g, s) \tag{21}$$
$$c_i = g e^{\sigma\epsilon} + n_i(g, s) - b$$

Then, we update our guess

$$\hat{v}^{nd}(b, g, s, \epsilon) = \max_i \left\{\hat{v}_i^{nd}(b, g, \epsilon) \ \text{ s. to } \ n_i \in \mathcal{S}(g, s)\right\} \tag{22}$$

When computing the max in equation (22), we need to impose that the issuance choice is within the schedule that corresponds to the sunspot s. In practice, when $s = s_b$, we impute a very negative pay-off for choices that are not in the schedule.

Compare $\hat{v}^{nd}(b, g, s, \epsilon)$ to the initial guess $v^{nd}(b, g, s, \epsilon)$. If it's close, we are done. Otherwise, update $v^{nd}(b, g, s, \epsilon)$ and go back to *Step 2*.